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Credit - Drishti Ias

Mombasa Declaration Adopted to Combat Illegal Fishing

Source: AP

The **Mombasa Declaration was adopted at the 11th Our Ocean Conference in Mombasa, Kenya,** by countries from Africa, Asia, Europe, the Caribbean and the Pacific to strengthen action against **illegal, unreported and unregulated (IUU) fishing.**

- **Coalition for Fisheries Transparency: It was developed with support from the Coalition for Fisheries Transparency, a global network of over 60 civil society organisations working to improve transparency and accountability in fisheries governance and management.**
- **About the Declaration:** It is a call to action for **coastal and flag States to improve fisheries transparency,** especially through better collection, sharing and public access to information related to fishing vessels and fishing activities.
- **Governance Mechanism:** It supports the **Global Charter for Fisheries Transparency** , which outlines **10 low-cost policy principles** aimed at strengthening fisheries governance and ensuring sustainable marine resource management.
- **Key Commitments:** Signatory countries agreed to **introduce transparency reforms such as modernising vessel registries, publishing fishing authorisations, improving vessel ownership data** and strengthening information-sharing mechanisms.
- **Participating Countries:** Countries such as Belgium, Cameroon, Chile, Ghana, Guinea, Liberia, Panama, Papua New Guinea, Peru, Somalia, South Korea, France, The Gambia, Dominican Republic, Republic of the Congo and others endorsed the declaration, showing broad global support. (**India is not a signatory**).
- **Need:** IUU fishing threatens marine ecosystems, coastal communities, small-scale fishers, food security and livelihoods, especially in **developing and low-income countries** dependent on fisheries.
- **Impact:** IUU fishing is estimated to cost the global economy up to **$50 billion annually, while also distorting markets,** reducing income for legitimate fishers and contributing to declining fish stocks.
- **Concerns:** The issue is linked to serious **human rights violations** , including **forced labour, exploitation, and unsafe working conditions** in parts of the global fishing industry.
- **Significance:** Countries that signed the Mombasa Declaration are expected to begin implementing its commitments, while a wider campaign will encourage more nations to join before the next **Our Ocean Conference in 2027,** making transparency a global norm in fisheries governance.

Read More: WTO Agreement on Fisheries Subsidies

Sarvam Becomes India's Newest AI Unicorn

Source: FE

Bengaluru-based **startup Sarvam AI** has achieved **unicorn status** , becoming India's first dedicated full-stack **Artificial Intelligence (AI)** startup to be valued at over USD 1 billion. The milestone marks a significant boost for India's deep-tech and sovereign AI ecosystem.

- A "unicorn" refers to a privately held startup valued at over USD 1 billion. As of **June 2026, India has 131 unicorns,** making it the **world's third-largest unicorn ecosystem after the US and China** , with recent additions such as **Sarvam AI, Neysa, Skyroot, KreditBee, and Porter** highlighting a growing shift towards **deep-tech sectors, particularly AI and space technology** .
- **Sarvam AI** : It specializes in building India's **sovereign AI stack** . It focuses on reducing India's reliance on foreign AI platforms.
- **Indigenous Large Language Model (LLM) Mandate:** Under the aegis of the **Government of India's IndiaAI Mission** , Sarvam AI has been selected to develop the country's first indigenous LLM, constructing three distinct variants (**Sarvam-Large** for advanced reasoning, **Sarvam-Small** for real-time applications, and **Sarvam-Edge** for on-device operations) featuring up to **70-billion parameters** .
- **Unveiling of Frontier Models:** At the **India-AI Impact Summit 2026,** the startup unveiled two next-generation models **Sarvam-30B** (a 30-billion parameter model optimized for real-time conversations with a 32,000-token context window) and **Sarvam-105B** (a 105-billion parameter model with a 128,000-token context window tailored for complex multi-step reasoning).
- **Enterprise Platforms & Multilingual Tools:**
    - Sarvam Samvaad: Conversational AI agents that integrate with enterprise tools to generate insights and take actions using proprietary data.
    - Sarvam Dub: An AI dubbing model featuring zero-shot voice cloning and cross-lingual speech capability for seamless multilingual content creation.
    - Population-Scale Social Delivery: The startup introduced "Vikram" (a multilingual chatbot named after Vikram Sarabhai) and used its voice agents to collect high-quality data from 17 million farmers for the Ministry of Agriculture and Farmers Welfare.

Read more: Sarvam AI and the Sovereign AI in India

Supreme Court Declares Right to Walk a Fundamental Right

Source: TH

The Supreme Court of India delivered a landmark judgment in the case of ***Maniyar Iliyaz @ Shaik Riyaz v. P. Ayyappan & Ors. (2026)*** , declaring that the **right to walk on safe and demarcated footpaths is a Fundamental Right** .

- **Surge in Pedestrian Fatalities:** Contextualizing the urgency, data reveals a catastrophic rise in pedestrian vulnerabilities. **Pedestrian deaths surged by nearly 163% between 2015 and 2024** , with their share in total road fatalities more than doubling from **9.5% to 20.61%** .
- **Constitutional Basis:** The **right to walk is an integral part of the right to free movement guaranteed under Article 19(1)(d)** , and it also embodies expressional, congregational, and associational rights under **Articles 19(1)(a), 19(1)(b), and 19(1)(c), read in conjunction with the right to life under Article 21** .
- **Absolute Priority over Vehicles:** The Court held that while the **freedom to walk is subject to reasonable restrictions** , pedestrians' right to use designated footpaths takes precedence over the movement and **privileges of motorized vehicles, and public spaces cannot be monopolized by drivers.**
- **Enforceable Correlative Duty:** The mere existence of a road creates an **enforceable legal duty on public bodies, specifically Urban Development Authorities, Municipal Corporations, Municipalities, and Panchayats,** to demarcate, construct, and maintain safe footpaths for walkers.
    - Citizens now have the right to invoke **restitutionary constitutional and legal remedies against these duty-bearers for failing to provide safe footpaths** , a mechanism that is **completely independent of the compensation remedies available under the Motor Vehicles Act, 1988.**
- **Critique of the Motor Vehicles Act, 1988:** The Apex Court criticised the existing legislation, noting that it **undermines walkers' rights by establishing the "vehicle" as the primary subject** , treating human and pedestrian interests as merely incidental.
    - Emphasising the need for modern institutional governance, the Court urged the formulation of dedicated laws and the establishment of an **independent, full-time regulatory body** with domain expertise to protect and plan for pedestrian rights.
- **Suo Motu Intervention:** The Court directed its Registry to forward the judgment to the **Law Commission of India** and key ministries including **Housing and Urban Affairs, Rural Development, and Road Transport and Highways** to initiate the necessary legal framework and identify duty-bearers.

Read more: Right to Safe Travel on National Highways under Article 21

Biochar and its Applications

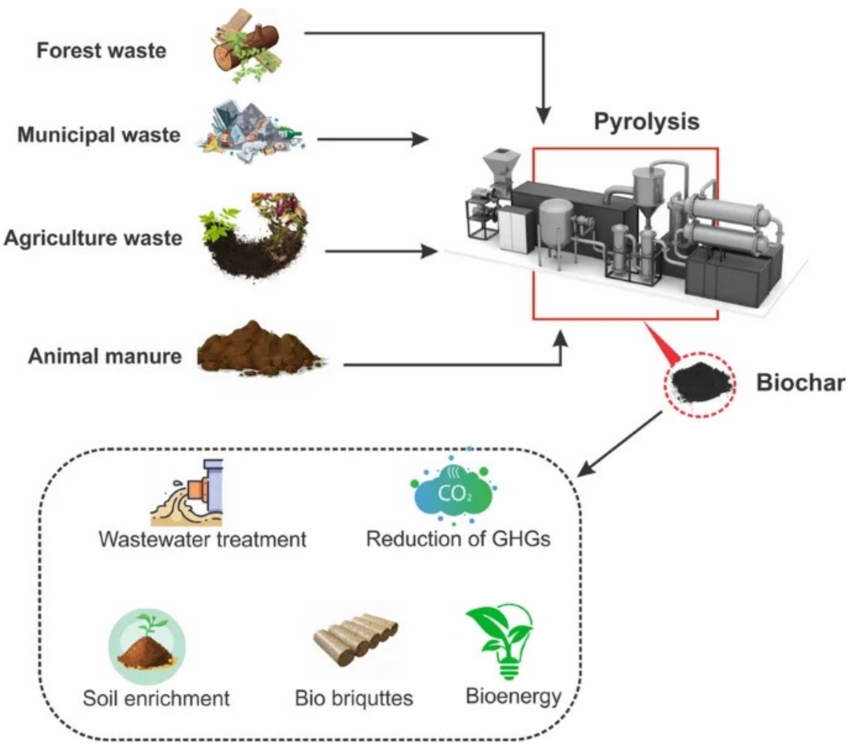
Source: TH

Why in News?

Biochar has emerged as a promising solution to two major challenges confronting Indian agriculture, namely large-scale crop residue burning and declining soil health. As India seeks climate-resilient and sustainable farming practices, biochar is gaining attention for its ability to **convert agricultural and organic waste into a valuable soil-enhancing, carbon-sequestering resource.**

What is Biochar?

- **Definition:** Biochar is a **highly porous, carbon-rich material** that resembles charcoal. It is produced through the thermal decomposition of organic biomass (such as agricultural waste, forestry residues, and food processing waste).
- **The Process (Pyrolysis):** It is generated through a process called **pyrolysis** , which involves heating the biomass in an oxygen-depleted environment at high temperatures (typically between **300°C and 700°C**).
- **Valuable By-products:** Along with solid biochar, the pyrolysis process also yields valuable by-products like **syngas** and **bio-oil** , which can be utilized as renewable energy sources.
- **Mechanism in Soil:** Its highly porous structure aggregates soil particles, effectively holding water and creating an ideal environment for beneficial microorganisms to thrive.



Significance and Applications

- **Enhancing Agricultural Productivity:** Applying biochar to degraded soils improves crop yields by **10% to 30%** , particularly in nutrient-deficient regions.

- It increases the soil's water-holding capacity by **10% to 25%** , helping crops withstand moisture stress during droughts and **heatwaves.**

- **Mitigating Air Pollution and Managing Waste:** India produces over **500 million tons** of crop waste annually. Converting **stubble into biochar provides a sustainable alternative to open-field burning in North India** , drastically reducing smoke and **particulate matter.**
  - It offers a solution for urban waste management by **converting biodegradable municipal solid garbage (which accounts for over 50% of India's 62 million tonnes of annual urban waste)** and sewage sludge, thereby diverting it from landfills.
- **Carbon Sequestration and Climate Action:** Biochar locks atmospheric carbon into a stable form that persists in the soil for centuries, preventing its release as **methane** or CO 2 .
  - One ton of biochar can sequester **2.5 to 3 tons of CO 2 equivalent** , making it a powerful nature-based climate solution.
- **Economic Viability via Carbon Credits:** Internationally recognized as a persistent carbon dioxide removal technology, each tonne of certified biochar can generate **2 to 2.8 tonnes of CO2-equivalent in carbon credits** .
  - This creates a lucrative alternative income stream for rural communities, cooperatives, and farmers who can sell it as a soil amendment, biofuel, or trade the carbon credits.
- **Power Generation:** Biochar production through pyrolysis in India can generate **approximately 20–30 million tonnes of syngas and 24–40 million tonnes of bio-oil** as valuable byproducts.
  - Syngas can produce 8–13 Terawatt-hour (TWh) of electricity annually.
  - Bio-oil can substitute 12–19 million tonnes of diesel or kerosene, helping reduce crude oil imports and cutting fossil fuel emissions by over 2%.
- **Water Purification:** Chemically modified **pristine biochar** can be utilized in wastewater treatment to efficiently filter out toxic heavy metals such as chromium, arsenic, and lead.
- **Construction Sector:** Incorporating 2-5% biochar into concrete improves mechanical strength, increases heat resistance by 20%, and captures approximately 115 kg of CO_2 per cubic metre.

Domestic and Global Success Stories

- **India:** Field trials in the Akola district of Maharashtra using maize stalk biochar **improved overall soil fertility and organic carbon content in black soils.**
  - Research in Kerala showed that biochar derived from coconut leaf stalks significantly increased soil quality across different cropping systems.
  - The **KISAN kiln** project from IIT-Kharagpur is actively allowing smallholders to monetise their farm waste.
- **Global Examples:**
  - **Kenya:** Converting rice husks into biochar has produced thousands of certified carbon credits while improving soil phosphorus content and pH.
  - **Thailand:** The government has tied **biochar certification to its national carbon registry system** , driving use through soil rehabilitation initiatives.
  - **Brazil:** Reported large yield gains and high carbon retention using biochar generated from sugarcane bagasse.

Frequently Asked Questions (FAQs)

1. What is biochar and how is it produced?
Biochar is a porous, carbon-rich material produced by heating biomass in an oxygen-limited environment through pyrolysis, typically at 300°C–700°C.

2. How does biochar help address crop residue burning?
It converts agricultural residues into a valuable soil amendment, reducing open-field burning, air pollution, and particulate emissions.

3. Why is biochar considered a climate mitigation tool?
Biochar locks carbon in a stable form for centuries and can sequester about 2.5–3 tonnes of CO_2 equivalent per tonne produced.

4. How can biochar generate economic benefits for farmers?
Farmers can earn income through improved crop yields, sale of biochar products, and participation in carbon credit markets.

5. What are the major by-products of the pyrolysis process?
Pyrolysis produces syngas and bio-oil alongside biochar, both of which can be utilized as renewable energy sources.

6. What role can biochar play in urban waste management?
Biochar can be produced from biodegradable municipal waste and sewage sludge, reducing landfill burden and promoting waste-to-wealth initiatives.

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[Watch Video on YouTube: ▶ https://www.youtube.com/embed/owuX0EUq3go]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Q. What is the use of biochar in farming? (2020)

1. Biochar can be used as a part of the growing medium in vertical farming.
2. When biochar is a part of the growing medium, it promotes the growth of nitrogen-fixing microorganisms.
3. When biochar is a part of the growing medium, it enables the growing medium to retain water for a longer time.

Which of the statements given above is/are correct?

(a) 1 and 2 only
(b) 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

Ans: (d)

Q. With reference to the usefulness of the by-products of the sugar industry, which of the following statements is/are correct? (2013)

1. Bagasse can be used as biomass fuel for the generation of energy.
2. Molasses can be used as one of the feedstocks for the production of synthetic chemical fertilizers.
3. Molasses can be used for the production of ethanol.

Select the correct answer using the codes given below:

(a) 1 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Ans: (c)